

Calculus and Analytical Geometry

Derivatives

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Objectives

After preparing this topic, you should be able to

1. Understand first derivative physically and explain it.
2. Understand second derivative physically and explain it.

The Why Section

- Suppose a machine pushes a component through a nonlinear spring, and the force required is modeled by

$$F(x) = 20x + 5x^2 - 0.5x^3$$

- In the above model, x is the displacement of the machine's actuator in meters and $F(x)$ is the force applied by the machine in newtons.
- At $x = 2$, what will be the rate of change of force i.e., near $x = 2$ m, if the displacement changes by a small amount δx , how much force changes?
- Will the force be increasing or decreasing at $x = 2$?
- If the force is increasing, is it increasing at increasing rate or decreasing rate?
- Similarly, if the force is decreasing, is it decreasing at increasing rate or decreasing rate?
- In order to understand it, we understand the derivatives physically in the foregoing discussion.

Derivatives in General

- Consider the general function

$$y = f(x) \quad (1)$$

- A point (x, y) is shown on the graph.

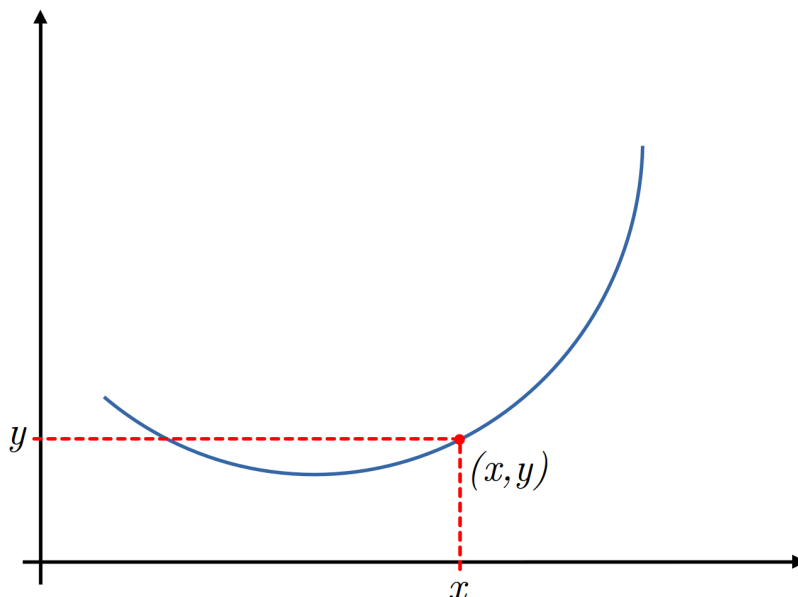
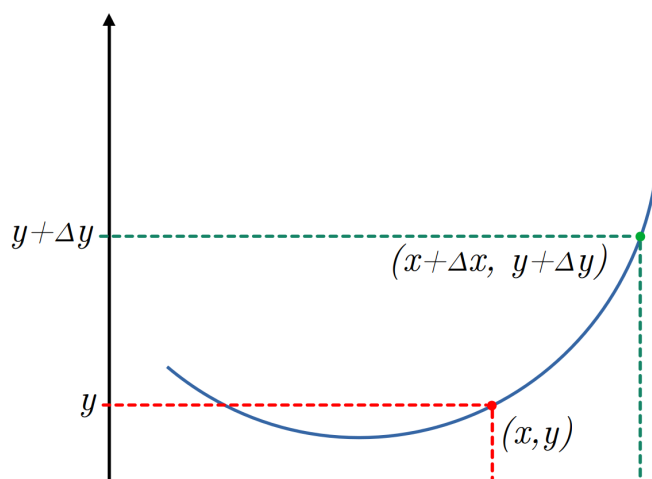


Figure 1. Graph of $y = \sin x$ and the point (x, y) on the graph.

- We are interested in finding the instantaneous change in y with respect to x .
- Before looking into the instantaneous change, we look into the change in y as x changes.
- This change can be computed as shown below.

$$y + \Delta y = f(x + \Delta x) \quad (2)$$



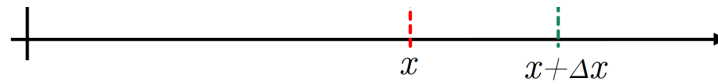


Figure 2. Graph of $y = \sin x$ with two points shown on the graph.

- We are interested in finding the change in y (Δy) per unit change in x (Δx) *i.e.*, how much y changes per unit change in x ($\frac{\Delta y}{\Delta x}$).
- Therefore, we modify (2) as shown below.

$$\Delta y = f(x + \Delta x) - y \quad (1)$$

$$= f(x + \Delta x) - f(x) \quad (\text{Using equation (1)}) \quad (2)$$

- In order to arrive at $\frac{\Delta y}{\Delta x}$, we divide both sides of above equation.

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

- We can modify the denominator of the above equations as shown below.

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{(x + \Delta x) - x} \quad (3)$$

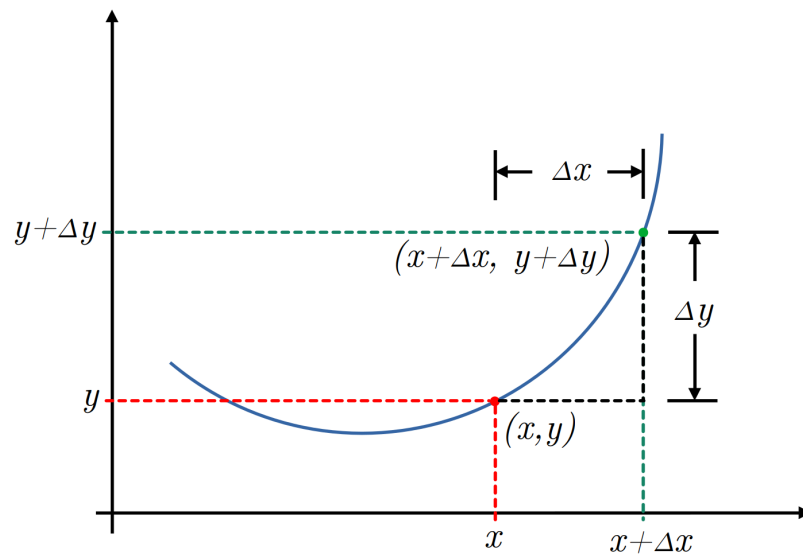


Figure 3. Graph of $y = \sin x$ with changes Δx and Δy .

- Let

$$y_1 = f(x) \quad (3)$$

$$y_2 = f(x + \Delta x) \quad (4)$$

$$x_1 = x \quad (5)$$

$$x_2 = x + \Delta x \quad (6)$$

- Using the above four relations, we can write equation (3) as follows.

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{(x + \Delta x) - x} = \frac{y_2 - y_1}{x_2 - x_1} \quad (4)$$

- It means, equation (3) represents the slope of the line.
- Note that the points on the right hand side of equation (3) are the coordinates of points P and Q in figure 2.
- These points intersect at two different points on the graph, therefore the line passing through these points is the secant line and the slope of the line in equation (3) or (4) is the slope of the secant line.

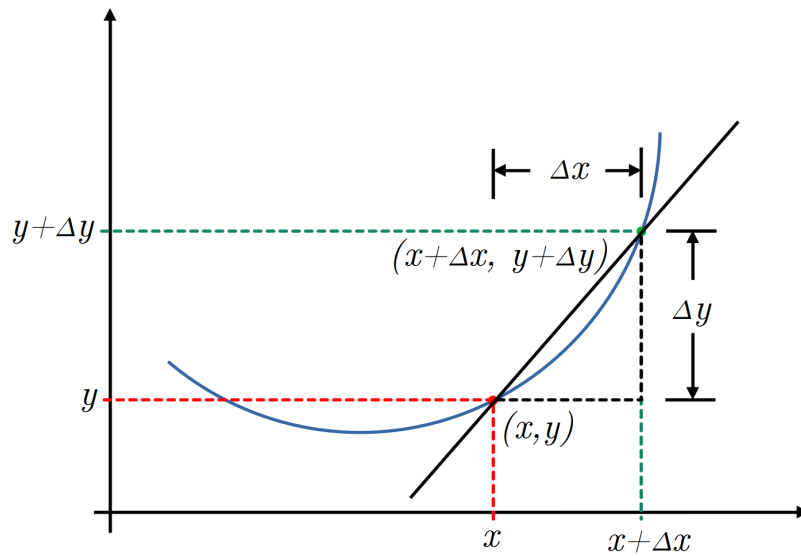


Figure 4. Graph of $y = \sin x$ and the secant line through two points.

- In real physical world, we deal with problems involving continuously varying quantities and in these problems we need instantaneous changes in the dependent variable with respect to (per unit change) the independent variable.
- Let's represent the small change with δ and this change is so small that $\delta \rightarrow 0$.
- Therefore, we modify equation (3) as shown below.

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad (5)$$

- Since $\delta x \rightarrow 0$, it means the points x and $x + \delta x$ are very close such that

$$(x + \delta x) - x \approx 0$$

- It confirms that point P approaches point Q and therefore secant line becomes the tangent line as shown in figure.
- This gives the instantaneous change.
- In short form we write

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$$

- So, finally we can write the slope of tangent line (equation (5)) as given below.

$$\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad (6)$$

- Suppose the angle between the tangent line with the horizontal is θ , then the angle between the tangent line and the line δx is also θ , however the triangle is a micro triangle.
- In the micro triangle, we have the following trigonometric identity.

$$\tan \theta = \frac{f(x + \delta x) - f(x)}{\delta x}$$

- Since $\delta x \rightarrow 0$, we can rewrite the above equation as shown below.

$$\tan \theta = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad (7)$$

- Equations (6) and (7) confirm the following identity.

$$\tan \theta = \frac{dy}{dx}$$

- This is because, $\tan \theta$ is also the slope of line (hypotenuse).
- Finally, equation (6) says that we can find derivative along the curve, but in real world problems we need derivatives at fixed points *e.g.*, at $t = 5$ or $x = 3$ etc.

Derivatives in Real World

The First Derivative

- Consider the following simple trigonometric function.

$$y = \sin x$$

- The first derivative of this function is

$$\frac{dy}{dx} = \cos x$$

- Instead of using the first principle, we are interested in derivatives at fixed points.
- For a complete understanding, we consider the following points.

$$x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$$

Case-I: $x = \frac{\pi}{4}$

- For $x = \pi/4$,

$$y = \sin(\pi/4) = 0.707$$

- So, the point is $(x, y) = (\pi/4, 0.707)$.

- Now,

$$m = \frac{dy}{dx} = \cos x = \cos(\pi/4) = 0.707$$

- Using the point $(\pi/4, 0.707)$ slope (0.707) form, we can find the y -intercept as given below.

$$y - y_1 = m(x - x_1)$$

$$y - 0.707 = 0.707(x - \pi/4)$$

- On y -axis, $x = 0$, therefore

$$y - 0.707 = -0.707(\pi/4)$$

$$y = 0.707 - 0.707(\pi/4) = 0.1513 = c$$

- Now, we can write the equation of line.

$$y = 0.707x + 0.1513$$

- At $x = 0$, $y = 0.1513$ and at $x = \pi/2$, $y = 1.262$.
- Therefore, we have two points $(x_1, y_1) = (0, 0.1513)$ and $(x_2, y_2) = (\pi/2, 1.262)$.
- Joining these two points will automatically draw the tangent to the graph of $y = \sin x$ at $x = \pi/4$.

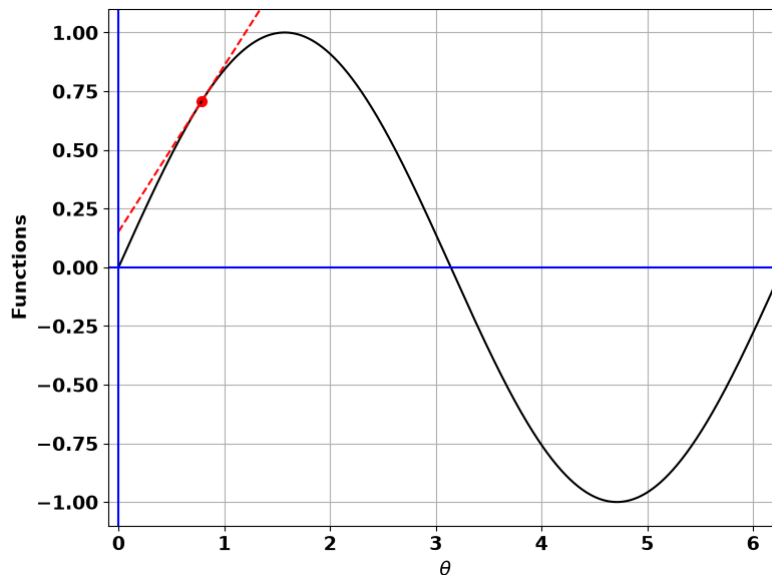


Figure 5. Tangent to $y = \sin x$ at $x = \pi/4$.

- A Python program to draw tangent line at any point on the graph of $y = \sin x$ is shown below.

```
import numpy as np
import matplotlib.pyplot as plt

def func(x):
    return np.sin(x)

def dydx(x):
    return np.cos(x)

def newy(x, m, c):
    return m*x + c

x = np.linspace(0, 2*np.pi, 1000)
y = func(x)
```

```

x1 = np.deg2rad(float(input("Enter the angle (in degrees):
                             ")))
y1 = func(x1)

m = dydx(x1)

c = y1 - m*x1

xl = x1 - np.pi/4
xh = x1 + np.pi/4

yl = newy(xl, m, c)
yh = newy(xh, m, c)

plt.figure(figsize=(8, 6))
plt.plot(x, y, '-k')
plt.plot([xl, xh], [yl, yh], '--r')
plt.plot([-0.10, 2*np.pi], [0, 0], '-b')
plt.plot([0, 0], [-1.1, 1.1], '-b')
plt.scatter(x1, y1, s=50, c='r')
plt.xlim([-0.10, 2*np.pi])
plt.ylim([-1.1, 1.1])
plt.xlabel(r'$\theta$', size=14, weight='bold')
plt.ylabel('Functions', size=14, weight='bold')
plt.xticks(size=14, weight='bold')
plt.yticks(size=14, weight='bold')
plt.grid(axis='both')
plt.tight_layout()
plt.show()

```

- We see that the slope of the tangent line at $x = \pi/4$ is $m = 0.707$.
- We want to confirm this slope using $\tan \theta$.
- We can find θ using dot product.
- Let's first find out the vectors.
- One point on the tangent line is the y -intercept and the coordinates of this point are $(0.0, 0.1513)$.

- The second point on the tangent line is $(x_2 = \pi/2, y_2 = 0.707 * \pi/2 + 0.1513) = (1.5708, 1.2619)$.
- The vector, say **a** formed by these two points is $\mathbf{a} = \langle 1.5708 - 0.0, 1.2619 - 0.1513 \rangle = \langle 1.5708, 1.1106 \rangle$.
- The vector **b** can be drawn with $(0, 0)$ and any point on x -axis *e.g.*, $(\pi/2, 0.0) = (1.5708, 0.0)$.
- The vector formed by these two points is $\mathbf{b} = \langle 1.5708 - 0, 0.0 - 0.0 \rangle = \langle 1.5708, 0.0 \rangle$.
- Using the two vectors $\mathbf{a} = 1.5708\mathbf{i} + 1.1106\mathbf{j}$ and $\mathbf{b} = 1.5708\mathbf{i} + 0\mathbf{j}$, we can find the angle between these two vectors (tangent line and x -axis) using dot product.
- So,

$$\theta = \cos^{-1} \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

- So, we get

$$\theta = \cos^{-1} \frac{(1.5708\mathbf{i} + 1.1106\mathbf{j}) \cdot (1.5708\mathbf{i} + 0\mathbf{j})}{\sqrt{1.5708^2 + 1.1106^2} \sqrt{1.5708^2 + 0^2}} = 0.615427 \text{ radian}$$

- Finally, we get the slope using **tan** function as below.

$$m = \tan(0.615427) = 0.707$$

- We see that slope obtained from derivative and trigonometry are the same.
- We can draw the tangent line as already discussed.
- The Python program to compute angle between the given vectors and then computing slope using **tan** θ is given below.

```
import numpy as np
```

```
m = np.cos(np.pi/4)
c = 0.1513
```

```

x1 = 0.0
y1 = m*x1 + c

x2 = np.pi/2
y2 = m*x2 + c

pi = np.array([x1, y1])
pf = np.array([x2, y2])

a = pf - pi

b = np.array([np.pi/2, 0.0])

adotb = np.dot(a, b)

am = np.linalg.norm(a)

bm = np.linalg.norm(b)

term = adotb/(am*bm)

theta = np.arccos(term)

print(np.tan(theta))

```

Case-II: $x = \frac{\pi}{2}$

- For $x = \pi/2$, $y = \sin \pi/2 = 1$.
- The slope at $x = \pi/2$ is

$$m = \cos \pi/2 = 0$$

- The y -intercept is

$$y = mx + c \implies y = c = 1$$

- Therefore the equation of tangent line at $x = \pi/2$ is

$$y = 1$$

- The slope at $x = \pi/2$ is shown below.

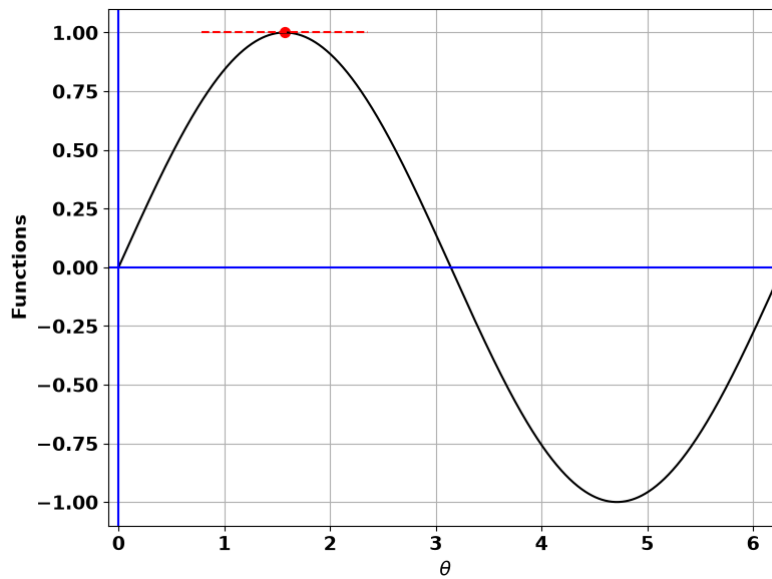


Figure 6. Tangent to $y = \sin x$ at $x = \pi/2$.

Case-III: $x = \frac{3\pi}{2}$

- Calculate yourself.

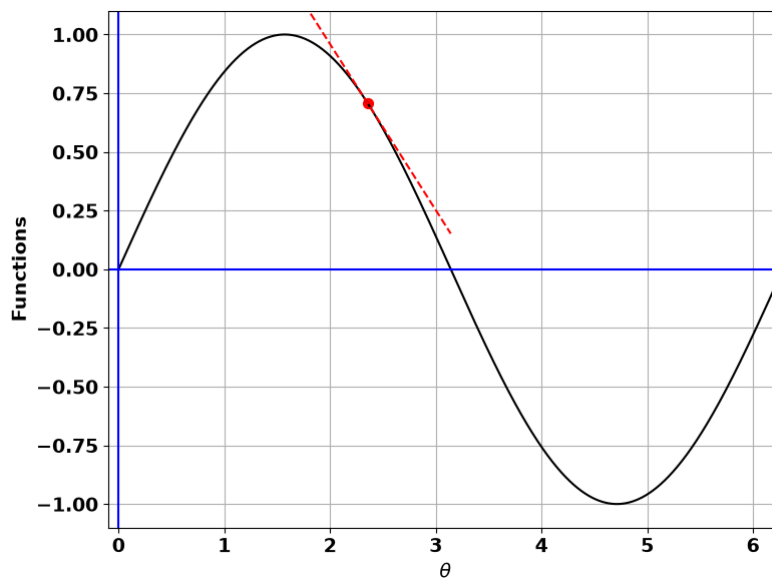


Figure 7. Tangent to $y = \sin x$ at $x = 3\pi/4$.

Case-IV: $x = \frac{5\pi}{4}$

- Calculate yourself.

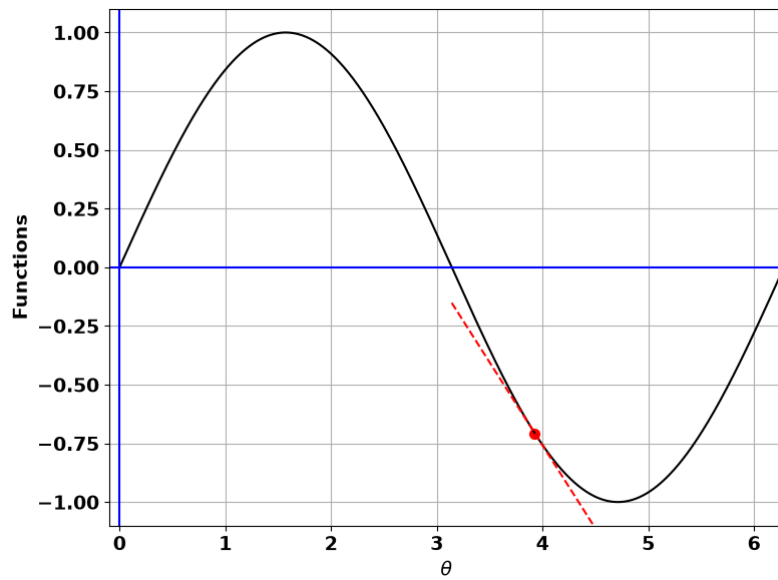


Figure 8. Tangent to $y = \sin x$ at $x = 5\pi/4$.

Case-V: $x = \frac{3\pi}{2}$

- Calculate yourself.

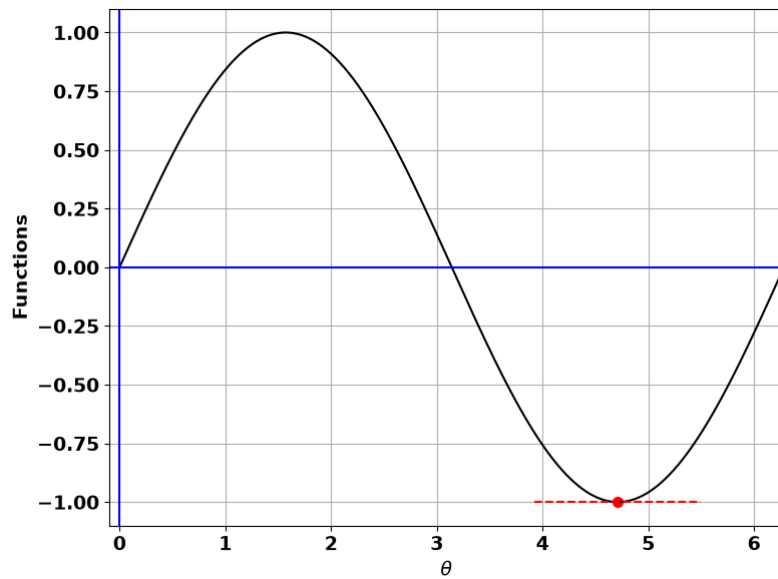


Figure 9. Tangent to $y = \sin x$ at $x = 3\pi/2$.

Case-VI: $x = \frac{7\pi}{4}$

- Calculate yourself.



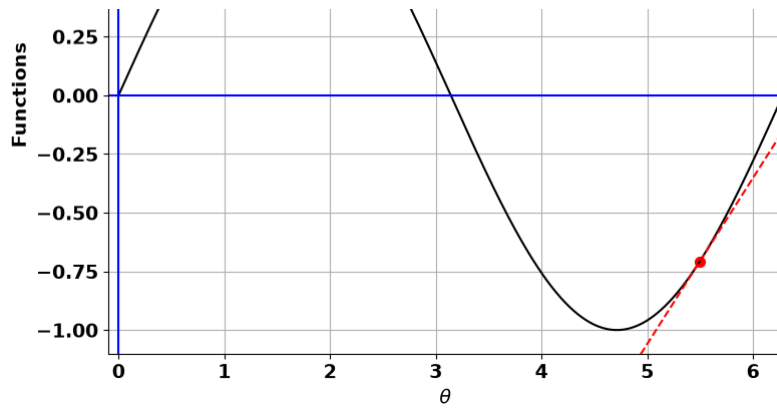


Figure 10. Tangent to $y = \sin x$ at $x = 7\pi/4$.

Shape of Curve with First Derivative

- The slope of tangent line gives the idea of the shape of curve at a point.

Case-A: Positive Slope

- If the slope of tangent line is positive, the curve may have one of the following two shapes.

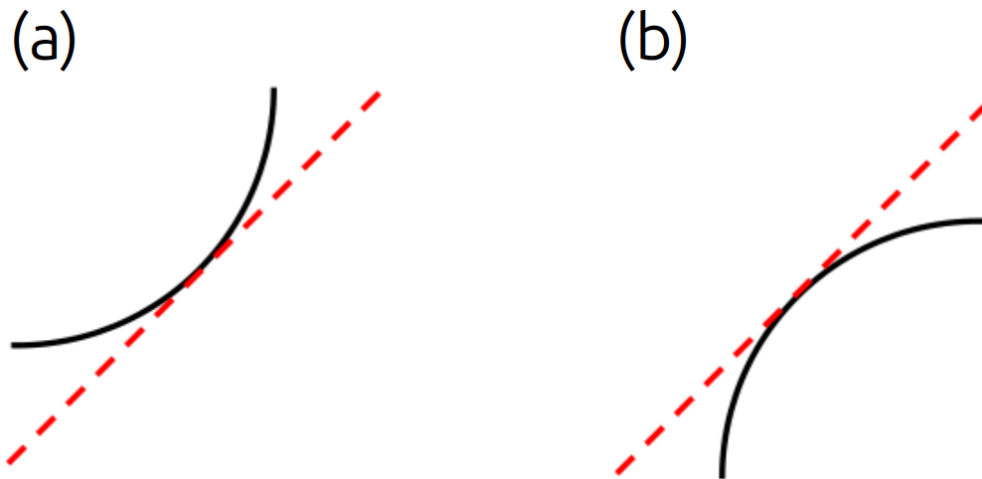


Figure 11. Shape of curve having positive slope.

Case-B: Negative Slope

- If the slope of tangent line is negative, the curve may have one of the following two shapes.



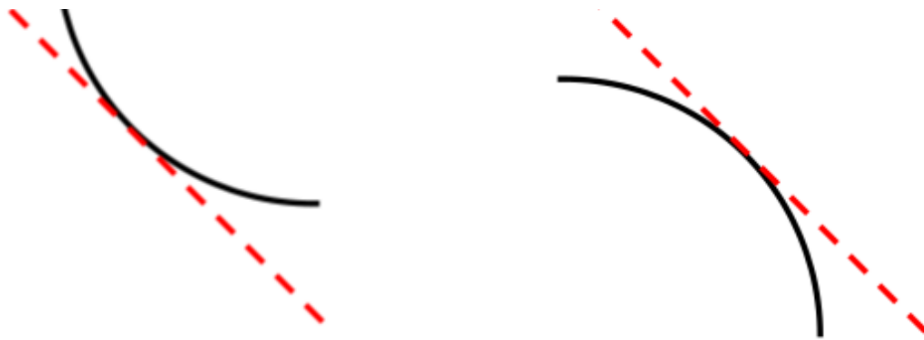


Figure 12. Shape of curve having negative slope.

- Consider the standard normal distribution function

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \quad (8)$$

- What is the slope of tangent line at $x = -2.0$?
- In order to find the slope of tangent line to the curve of equation (8), we find the derivative of $f(x)$ at $x = -2.0$.

$$f'(x) = -\frac{x}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \quad (9)$$

$$f'(x)|_{x=-2.0} = -\frac{(-2.0)}{\sqrt{2\pi}} e^{\frac{-(-2.0)^2}{2}} = 0.1080$$

- The slope of tangent line to the curve of normal distribution function at $x = -2.5$ is positive (0.1080).
- What will be the shape of curve at $x = -2.5$?

The Second Derivative

- The second derivative of the standard normal distribution function is

$$f''(x) = \frac{x^2 - 1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \quad (10)$$

- This time, we calculate the second derivative at the following four points.

- $x = -2$
- $x = -0.4$
- $x = 0.4$

4. $x = 2$

$$f''(-2.0) = \frac{(-2.0)^2 - 1}{\sqrt{2\pi}} e^{\frac{-(-2.0)^2}{2}} = 0.1620$$

$$f''(-0.4) = \frac{(-0.4)^2 - 1}{\sqrt{2\pi}} e^{\frac{-(-0.4)^2}{2}} = -0.3093$$

$$f''(0.4) = \frac{(0.4)^2 - 1}{\sqrt{2\pi}} e^{\frac{-(0.4)^2}{2}} = -0.3093$$

$$f''(2.0) = \frac{(2.0)^2 - 1}{\sqrt{2\pi}} e^{\frac{-(2.0)^2}{2}} = 0.1620$$

- Looking at the tangents at the above four points, we conclude the following points.
 - If the second derivative is positive at some point $x = a$, the curve will lie above the tangent line or in other words, the tangent line will be below the curve.
 - If the second derivative is negative at some point $x = a$, the curve will lie below the tangent line or in other words, the tangent line will be above the curve.

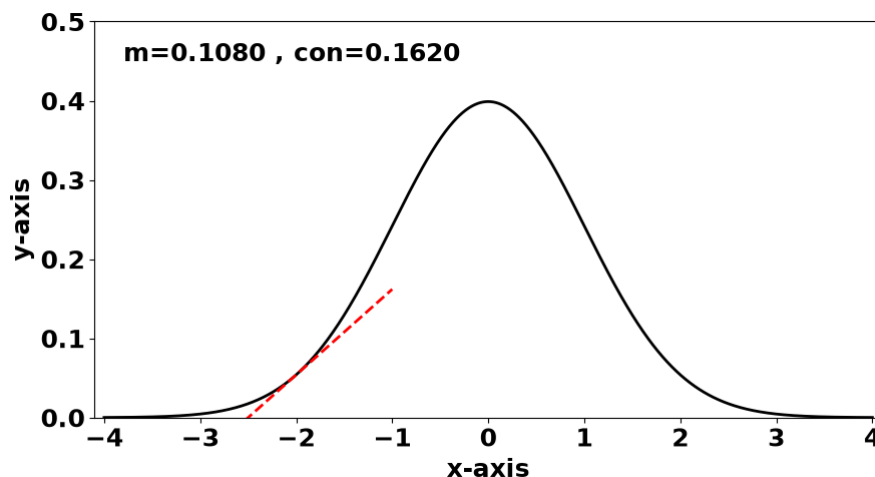
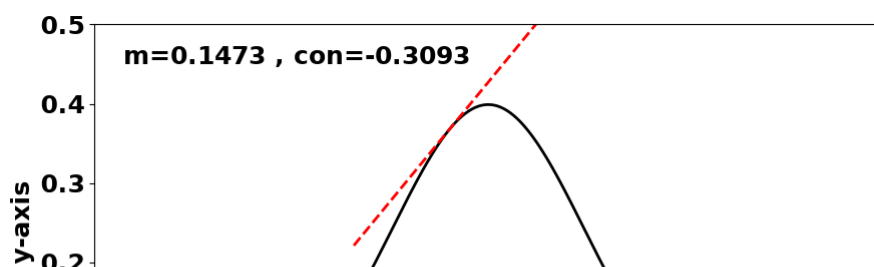


Figure 13. Slope and concavity at $x = -2$.



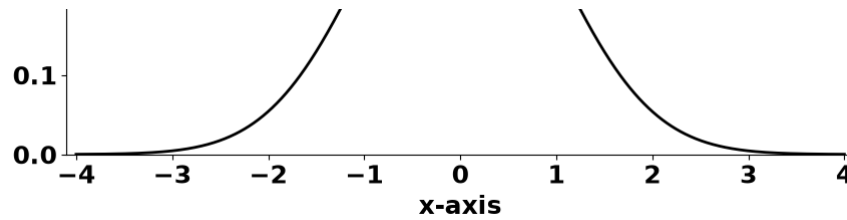


Figure 14. Slope and concavity at $x = -0.4$.

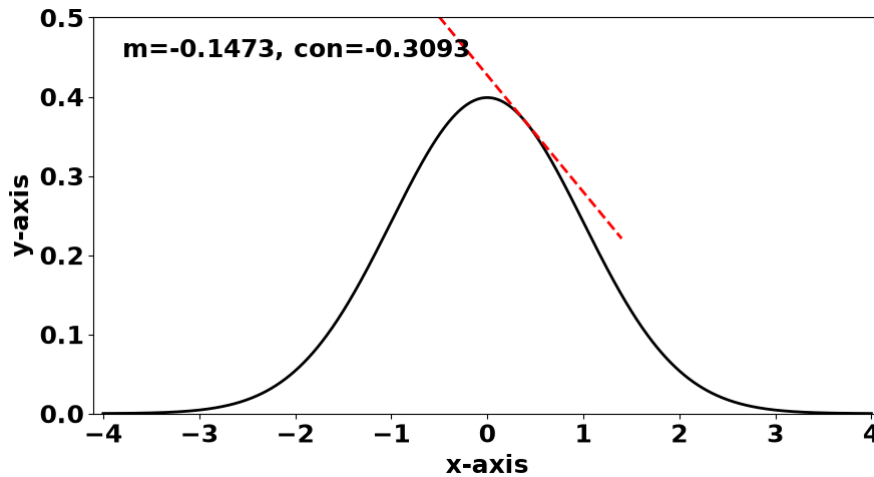


Figure 15. Slope and concavity at $x = 0.4$.

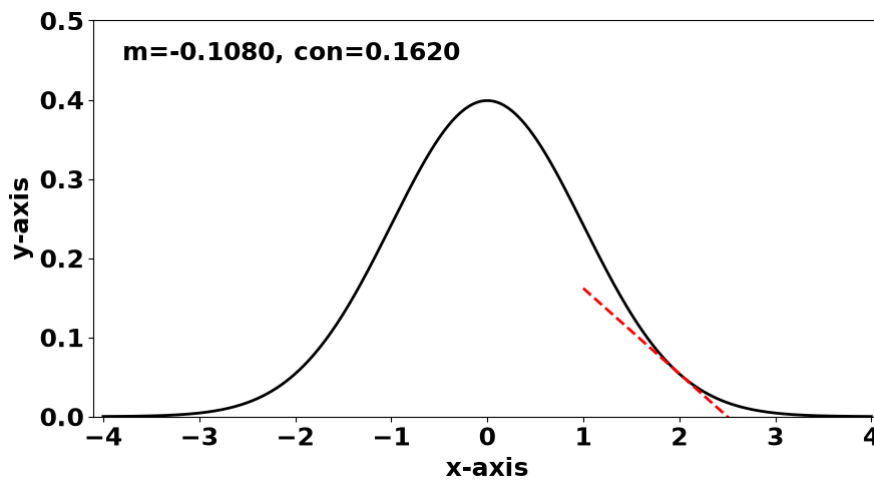


Figure 16. Slope and concavity at $x = 2$.

- The Python program used to draw figures 13 through 16 is as below.

```
import numpy as np
import matplotlib.pyplot as plt

def snd(x):
    return 1/(np.sqrt(2*np.pi))*np.exp(-x**2/2)
```

```

def sndd(x):
    return -x/(np.sqrt(2*np.pi))*np.exp(-x**2/2)

def snddd(x):
    return (x**2 - 1)/(np.sqrt(2*np.pi))*np.exp(-x**2/2)

x = np.linspace(-4, 4, 1000)
y = snd(x)

xps = np.array([-2.0, -0.4, 0.4, 2.0])
yps = snd(xps)
mps = sndd(xps)
cops = snddd(xps)
cps = yps - mps*xps
print(mps)
print(cops)

for i in range(len(xps)):
    plt.figure(figsize=(9, 5))
    plt.plot(x, y, '-k', linewidth=2)
    plt.plot(
        [xps[i]-1, xps[i]+1],
        [mps[i]*(xps[i]-1)+cps[i],
         mps[i]*(xps[i]+1)+cps[i]],
        '--r',
        linewidth=2
    )
    plt.xlabel("x-axis", size=18, weight='bold')
    plt.ylabel("y-axis", size=18, weight='bold')
    plt.xticks(size=18, weight='bold')
    plt.yticks(size=18, weight='bold')
    plt.xlim([-4.1, 4.1])
    plt.ylim([0, 0.5])
    plt.text(-3.8, 0.45, f"m={mps[i]:<7.4f},
        con={cops[i]:.4f}",
        size=18, weight='bold')
    plt.tight_layout()
    plt.savefig(f"slope_{i:d}.png")

```

Finalized Shape of Curve

- Now, using first and second derivatives, we can understand the shape of the curve at any point.

Case-I: $m > 0$, $\text{con} > 0$

- If $m > 0$ and $\text{Concavity} > 0$, the shape of the curve will be as shown below and the graph is said concave up (drawn up on the tangent).

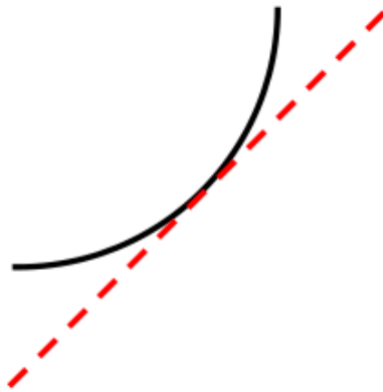


Figure 17. Shape of curve with $m > 0$ and $\text{Concavity} > 0$.

Case-II: $m > 0$, $\text{con} < 0$

- If $m > 0$ and $\text{Concavity} < 0$, the shape of the curve will be as shown below and the graph is said concave down (drawn down the tangent).



Figure 18. Shape of curve with $m > 0$ and $\text{Concavity} < 0$.

Case-III: $m < 0$, $\text{con} > 0$

- If $m < 0$ and $\text{Concavity} > 0$, the shape of the curve will be as shown

below and the graph is said concave up (drawn up on the tangent).

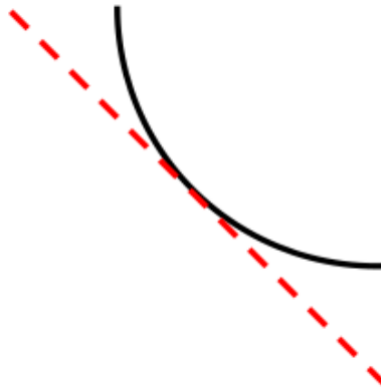


Figure 19. Shape of curve with $m < 0$ and Concavity > 0 .

Case-IV: $m < 0$, $\text{con} < 0$

- If $m < 0$ and Concavity < 0 , the shape of the curve will be as shown below and the graph is said concave down (drawn down the tangent).



Figure 20. Shape of curve with $m < 0$ and Concavity < 0 .

- Looking at the slopes and concavity of the standard normal distribution function, we conclude the following points.
 - At $x = -2$:
 - $f'(x) > 0 \implies$ height of $f(x)$ is increasing.
 - $f''(x) > 0 \implies$ the height is increasing rapidly.
 - At $x = -0.4$:
 - $f'(x) > 0 \implies$ height of $f(x)$ is increasing.

- $f''(x) < 0 \implies$ the height is increasing slowly.
- At $x = 0.4$:
 - $f'(x) < 0 \implies$ height of $f(x)$ is decreasing.
 - $f''(x) < 0 \implies$ the height is decreasing slowly.
- At $x = 2$:
 - $f'(x) < 0 \implies$ height of $f(x)$ is decreasing.
 - $f''(x) > 0 \implies$ the height is increasing rapidly.

Summary

- The first derivative represents the **instantaneous rate of change** of a dependent variable with respect to an independent variable and geometrically represents the **slope of the tangent line** to a curve.
- The derivative is obtained by considering the limit of the average rate of change as the change in the independent variable approaches zero, causing the secant line to approach the tangent line.
- The relationship between the derivative and the tangent angle is $(=)$, confirming that the derivative gives the slope of the tangent line.
- The first derivative can be evaluated at a fixed point to determine the slope and direction of a curve at that point; a positive derivative indicates an increasing function, while a negative derivative indicates a decreasing function.
- The sign of the first derivative alone does not completely determine the shape of a curve, because curves with the same positive or negative slope can have different shapes.
- The second derivative describes how the **slope of the curve changes** and determines its concavity: a positive second derivative corresponds to a curve lying above its tangent line, while a negative second derivative corresponds to a curve lying below its tangent line.
- Combining the signs of the first and second derivatives gives four possible local shapes: increasing and concave up, increasing and concave down, decreasing and concave up, and decreasing and concave down.
- For the standard normal distribution, the first derivative determines whether the function's height is increasing or decreasing, while the second

derivative indicates whether that increase or decrease is occurring rapidly or slowly.

- The overall discussion uses physical and graphical interpretations of the first and second derivatives to understand **slope, rate of change, and concavity at a specific point** rather than treating derivatives only as algebraic formulas.

Exercises

Exercise 1

Suppose a machine pushes a component through a nonlinear spring, and the force required is modeled by

$$F(x) = 20x + 5x^2 - 0.5x^3$$

where x is the displacement of the machine's actuator, in meters and $F(x)$ is the force applied by the machine, in newtons.

What is the behavior of force at $x = -5, 3, 6, 11$? In particular, determine whether the force is increasing or decreasing and whether its rate of change is increasing or decreasing at each point.

Exercise 2

The temperature of a cooling component is modeled by

$$T(t) = 25 + 75e^{-0.2t}$$

where t is time in minutes and $T(t)$ is temperature in °C.

Determine the behavior of the temperature at

$t = 1, 3, 6, 10$.

In particular, determine whether the temperature is increasing or decreasing and whether its rate of change is increasing or decreasing at each time.

Exercise 3

The position of a vehicle moving along a test track is modeled by

$$s(t) = 20 + 10 \sin \left(\frac{\pi t}{6} \right)$$

where t is time in seconds and $s(t)$ is position in meters.

Determine the behavior of the vehicle's position at

$$t = 1, 3, 5, 7.$$

In particular, determine whether the position is increasing or decreasing and whether its rate of change is increasing or decreasing at each time.

Exercise 4

The charge stored in a device during a charging process is modeled by

$$Q(t) = 5 \ln(t + 1) + 2$$

where t is time in seconds and $Q(t)$ is charge in coulombs.

Determine the behavior of the charge at

$$t = 1, 3, 8, 15.$$

In particular, determine whether the charge is increasing or decreasing and whether its rate of change is increasing or decreasing at each time.

Exercise 5

The water level in a tank is modeled by

$$h(t) = 10 + \frac{20t}{t + 4}$$

where t is time in hours and $h(t)$ is the water level in meters.

Determine the behavior of the water level at

$$t = 1, 4, 8, 12.$$

In particular, determine whether the water level is increasing or decreasing

and whether its rate of change is increasing or decreasing at each time.